Programming SQL Server 2019

Database Terms and Concepts and Introducing SQL Server

Topics

- Introduction to DDL command
- SQL Server Database Structure
- Creating database using simple command
- Creating databases using details information
- Viewing database information
- Altering a database

Homework:

A. Create a database using T-SQL with
- One primary data file of
  Size: 10MB
  Filegrowth: 10%
  Maxsize: 100MB
- One Log file of
  Size: 10MB
  Filegrowth: 10%
  Maxsize: 100MB

B. Change the size of data file to 20MB

C. Add a data file 'ESAD_data1' of size 5MB to the database with other default options

D. Rename the database to TraineeDB

Important Points:

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Summary:

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SQL Statements

- SQL is case-insensitive
- Every SQL statement end with ; (semi-colon), which optional
- We add GO after a single or a group of SQL statements
- GO denotes end of a batch, we will discuss later about SQL Batches

SQL Comment

- A single line comment starts with --. For example
  ```
  --Insert data into trainees table
  ```
- A multiline comment is enclosed with /* and */
  ```
  /*
          Comment
  */
  ```
- A multi-line comment must be within a single a batch. The following is invalid
  ```
  /*
  --
  GO
  --
  */
  ```

Running SQL in Query Editor

- When Press F5 or Click on Execute button, All the statements in the selected editor are executed
- When you press F5 with a set of statement selected, only selected statement will be executed
- Each time a query is executed, messages are shown on messages window. If any error occurred, you can track it seeing the error messages

Parsing Query

- Press Ctrl + F5 or click on the parse button (√ marked tool)
- SSMS will parse the query without executing it and show any error is there
- You should always track syntax errors before executing it

What is SQL?

- SQL (Structured Query Language) is a database language designed for managing data in relational database management systems

What is T-SQL?

- Known as Transact SQL
- T-SQL (Transact-SQL) is a version of SQL from Sybase and Microsoft that add several features to the Structured Query Language (SQL) including transaction control, exception and error handling, row processing, and declared variables.
- Microsoft's SQL Server and Sybase's SQL server support T-SQL statements

DDL & DML

- SQL statements primarily fall into two group:
- DDL (Data Definition Language): DDL allows creating, altering or removing database objects - like database itself, tables, stored procedures etc.
- There are three basic DDL
  → CREATE - used to create database objects
  → ALTER - used to modify database objects
  → DROP - used to delete database object
- DML (Data Manipulation Language): DML allows retrieving, storing, modifying or deleting data
- There are four basic DML statements
  → SELECT – to retrieve data from table
  → INSERT – to store data into a table
  → UPDATE – to modify data in a table
  → DELETE – to delete data from a table

SQL Database structure

- A SQL server is made of at least two files
- One data file (.mdf)
  → One transaction log file (.ldf)
  → Data File - It holds actual data
- Transaction log file - it does not hold data. It keeps track of data modification you make. If you change data SQL server first write what it is going to do in transaction log file and then it makes actual in data in data file.

Basic CREATE object Syntax

- In simplest form the syntax is
  ```
  CREATE object-type object-name
  ```
- Sometimes you provide extra information during creation of the object, the syntax for that
  ```
  CREATE object-type object-name
  (
          Extra information
  )
  ```

Let's create a database

- Open SSMS
- Click on "new query" tool
- Write the following
  ```
  CREATE database sqlStore
  GO
  ```
- Select the lines
- PRESS F5 or Execute button on toolbar
- You will see the following message in messages window

```
Command(s) completed successfully.
```

View Database information

- To extract information of an existing database, use sp_helpdb built-in procedure
- The syntax is
  ```
  execute sp_helpdb 'database-name'
  ```
- sp_helpdb is a built-in stored procedure. SQL provides a lot of such procedures. We will learn stored procedures later

Try this

```
EXECUTE sp_help_db 'sqlStore'
GO
```

- You will see result I showing a lot of information including data and log file

Deleting a database

- To delete an object, we use drop statement
- The syntax is

```
DROP object-type object-name
```

Try this

```
DROP database sqlStore
GO
```

- You will see

```
Command(s) completed successfully.
```

Creating database with file details

- When you create a database using simple create statement, two files, data and log files, are created at predefined location (in data folder under SQL server's installation directory with default values of other attributes.
- For default values, SQL Server depends model system database.
- But you can create database with files at location of your own choice and explicit values of various file attributes
- The syntax is:

```
CREATE database database-name
ON
(
        NAME = logical_file_name ,
        FILENAME = 'os_file_name',
        SIZE = size,
        MAXSIZE = max_size,
        FILEGROWTH = growth_increment
)
LOG ON
(
    NAME = logical_file_name ,
    FILENAME = 'os_file_name',
    SIZE = size,
    MAXSIZE = max_size,
    FILEGROWTH = growth_increment
)
```

- ON(...) - encloses data file specification
- LOG ON (..) - encloses log file specification
- NAME - Is the logical name used in SQL Server when referencing the file?
- FILENAME - Specifies the operating system (physical) file name. The path and file name used by the operating system when you create the file.
- SIZE-Is the initial size of the file. The kilobyte (KB), megabyte (MB), gigabyte (GB), or terabyte (TB) units can be used. The default is MB. Specify a whole number.
- MAXSIZE - Specifies the maximum size to which the file can grow. The kilobyte (KB), megabyte (MB), gigabyte (GB), or terabyte (TB) suffixes can be used. The default is MB. Specify a whole number. Can be UNLIMITED - the file grows until the disk is full.
- FILEGROWTH - Is the amount of space added to the file every time new space is required. The value can be specified in MB, KB, GB, TB, or percent (%). If a number is specified without an MB, KB, or % suffix, the default is MB.
- NAME and FILENAME are mandatory.
- Other attributes are optional.
- You can create database with multiple data and log files
- The syntax is:

```
CREATE database
ON (...),
(..)
LOG ON (..),
(..)
```

Create a database with details

Try this

```
CREATE database sqlStore
ON
(
        name = 'sqlStoredata_1',
        filename = 'F:\sqlStore_data_1.mdf',
        size = 2mb,
        maxsize = 50mb,
        filegrowth = 10%
)
LOG ON
(
        name = 'sqlStorelog_1',
        filename = 'F:\sqlStore_log_1.ldf',
        size = 2mb,
        maxsize = 50mb,
        filegrowth = 1mb
)
GO
EXEC sp_helpdb 'sqlStore'
GO
```

Modify file of an existing database

- The syntax is
- ALTER database

```
MODIFY FILE
(
  NAME = logical_file_name ,
  FILENAME = 'os_file_name',
  SIZE = size,
  MAXSIZE =  max_size,
  FILEGROWTH = growth_increment
)
```

- You must specify NAME
- Let's change the size of the data file of the sqlStore database

Try this

```
ALTER database sqlStore
MODIFY file
(
        name = 'sqlStoredata_1',
        filename = 'F:\sqlStore_data_1.mdf',
        size = 4mb
)
GO
EXEC sp_helpdb 'sqlStore'
GO
```

Adding file to an existing database

- The syntax is:

```
ALTER database
ADD FILE
(
        NAME = logical_file_name ,
        FILENAME = 'os_file_name',
        SIZE = size ,
        MAXSIZE =  max_size,
        FILEGROWTH = growth_increment
)
```

- You must specify NAME and FILENAME

Try this

```
ALTER database sqlStore
ADD file
(
    name = 'sqlStoredata_2',
    filename = 'F:\sqlStore_data_2.ndf',
    size = 2mb,
    filegrowth = 10%
)
GO
EXEC sp_helpdb 'sqlStore'
GO
```

Deleting a file from database

- The syntax is

```
ALTER database database-name
REMOVE file logical-name
```

- A file must be empty before it can be deleted

Try this

```
ALTER DATABASE sqlStore
REMOVE FILE sqlStoredata_2
GO
EXEC sp_helpdb 'sqlStore'
GO
```

Renaming a database

- To rename a database, you have to make sure that no one is using the database.
- You can rename in two ways
- Using ALTER statement
- The syntax is

```
ALTER database database-name
Modify Name = new-name
```

- Using sp_renamedb stored procedure
- The Syntax is

```
EXECUTE sp_renamedb 'old-name', 'new-name'
```

Try this

```
ALTER DATABASE sqlStore
MODIFY name = demo
GO
```

- Refresh databases node in object explorer
- You will see the new name

Again, try this

```
EXEC sp_renamedb 'demo', 'sqlStore'
GO
```

- Refresh databases node in object explorer
- You will see the database is changed to sqlStore